

ASTE-331a: Spacecraft Systems Engineering

Introduction to Space Systems

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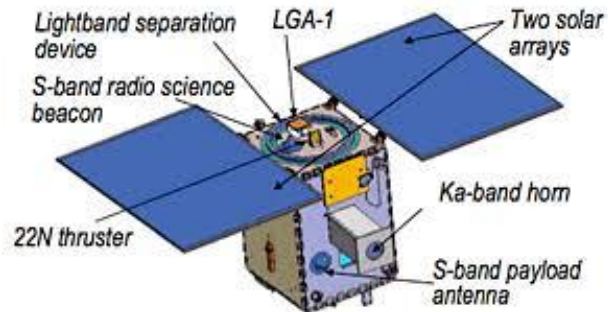
August 29th, 2025

- Introduction to Space Systems
- Case Study: GRAIL Conceptual Design
- Overview of Spacecraft Subsystems
- Mission Requirements Flowdown

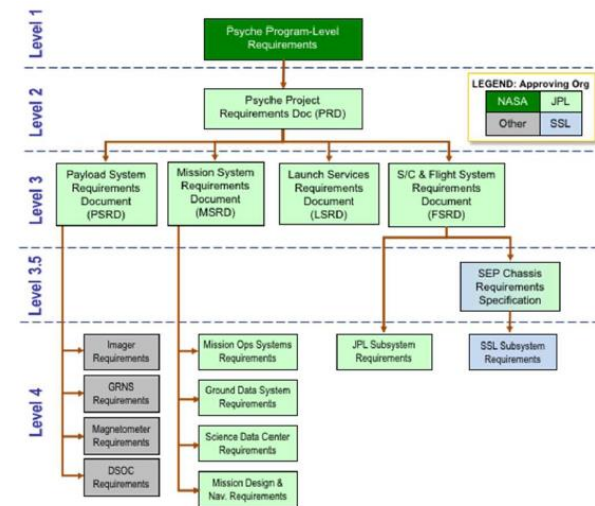
GRAIL Space System Concept



Preliminary Design



Requirements Flowdown



Near-Final Delivery



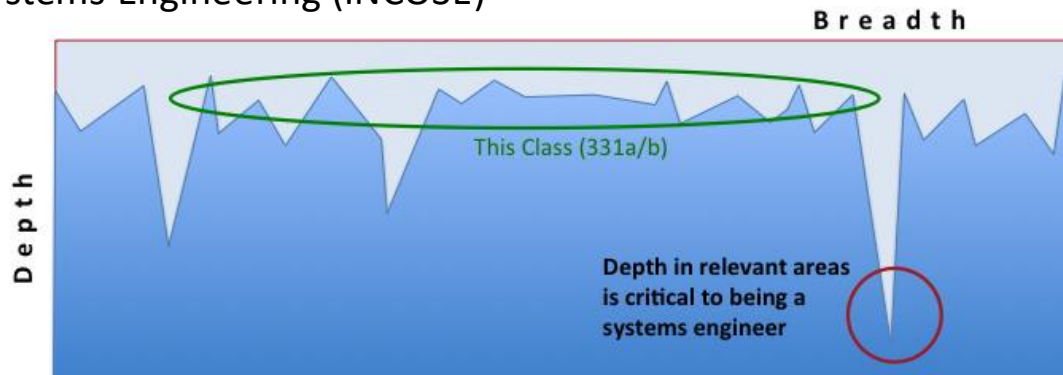
PART 1: SPACECRAFT SYSTEMS ENGINEERING

• Space Systems

- A system is a collection of “elements” that work synergistically together to perform one or more functions. –*NASA’s SE Handbook, 2017*
- “The planning, developing, integrating, testing, launching, and operation of **space systems** constitute a systems engineering effort of profound proportions. Such an undertaking requires close coordination of disparate participants, whose requisite scientific, engineering, and management capabilities span a breadth of disciplines that continue to change at dramatic rates.” – *Pisacane, 2011*

• Systems Engineers

- “Systems Engineers are at the heart of creating successful new systems. They are responsible for the system concept, architecture, and design. They analyze complexity and risk. They decide how to measure whether the deployed system actually works as intended. They are responsible for a myriad of other facets of system creation.” – International Council on Systems Engineering (INCOSE)



International Space Station



SpaceX Starship (first orbital test flight)

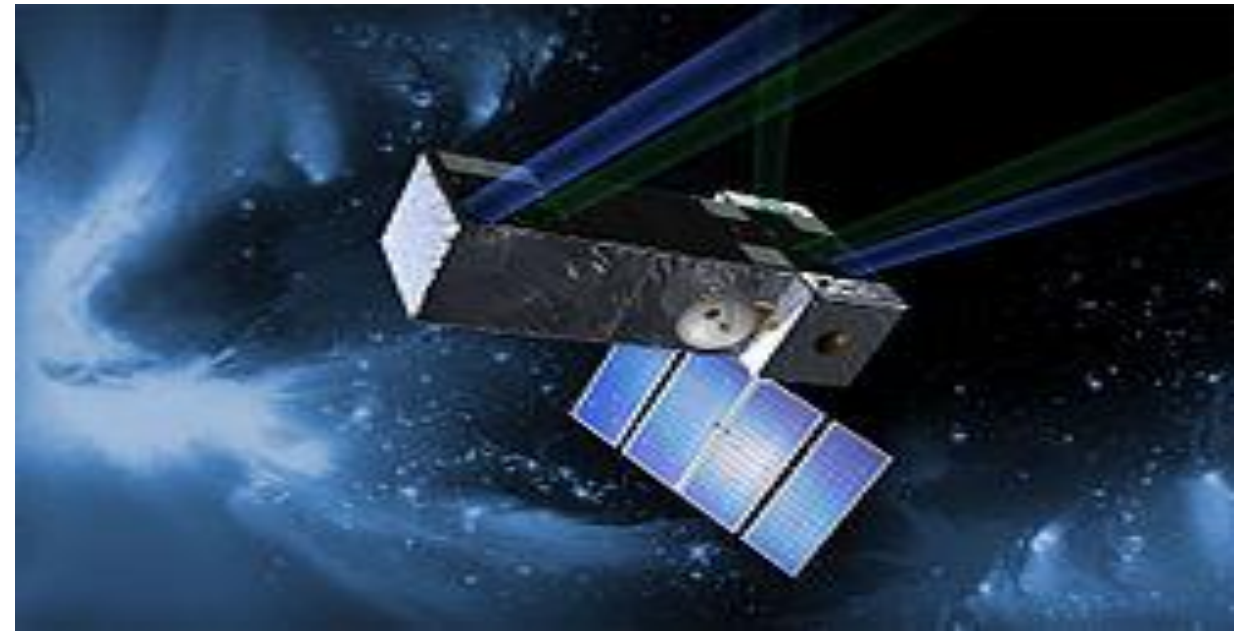


Examples (1 of 3)



- **STS-1 (Space Transportation System-1)** was the first orbital spaceflight of NASA's Space Shuttle program.
- The first orbiter, Columbia, launched on 12 April 1981 and returned on 14 April, 54.5 hours later, having orbited the Earth 36 times.

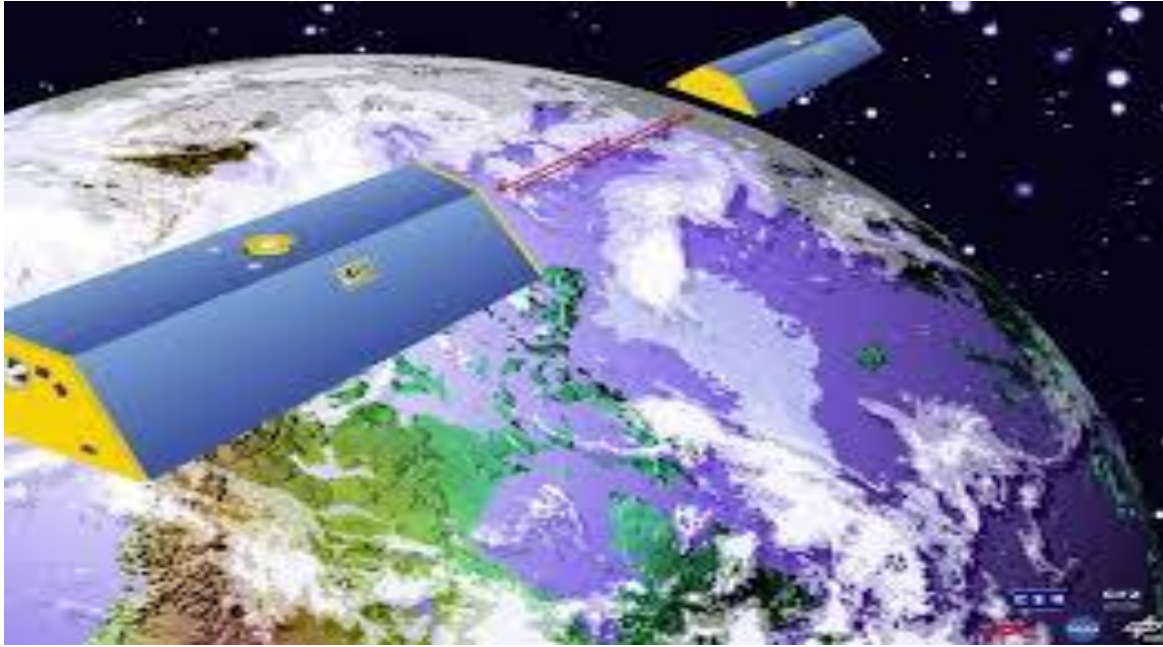
– *Wikipedia*



- **Space Interferometry Mission (SIM)** was a planned space telescope proposed by NASA.
- One of the main goals was the hunt for Earth-sized planets orbiting in the habitable zones of nearby stars.

– *Wikipedia*

Examples (2 of 3)



- The Gravity Recovery and Climate Experiment (GRACE) was a joint mission of NASA and the German Aerospace Center.
- Twin satellites took detailed measurements of Earth's gravity field anomalies from its launch in March 2002 to the end of its science mission in October 2017.

– *Wikipedia*



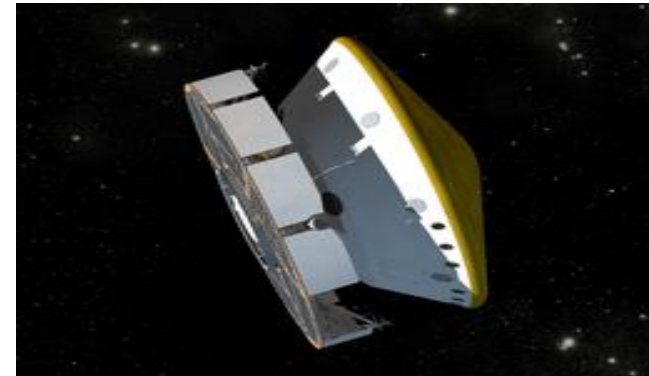
- The Gravity Recovery and Interior Laboratory (GRAIL) was a lunar science mission that used high-quality gravitational field mapping of the moon to determine its interior structure.

– *Wikipedia*

Examples (3 of 3)

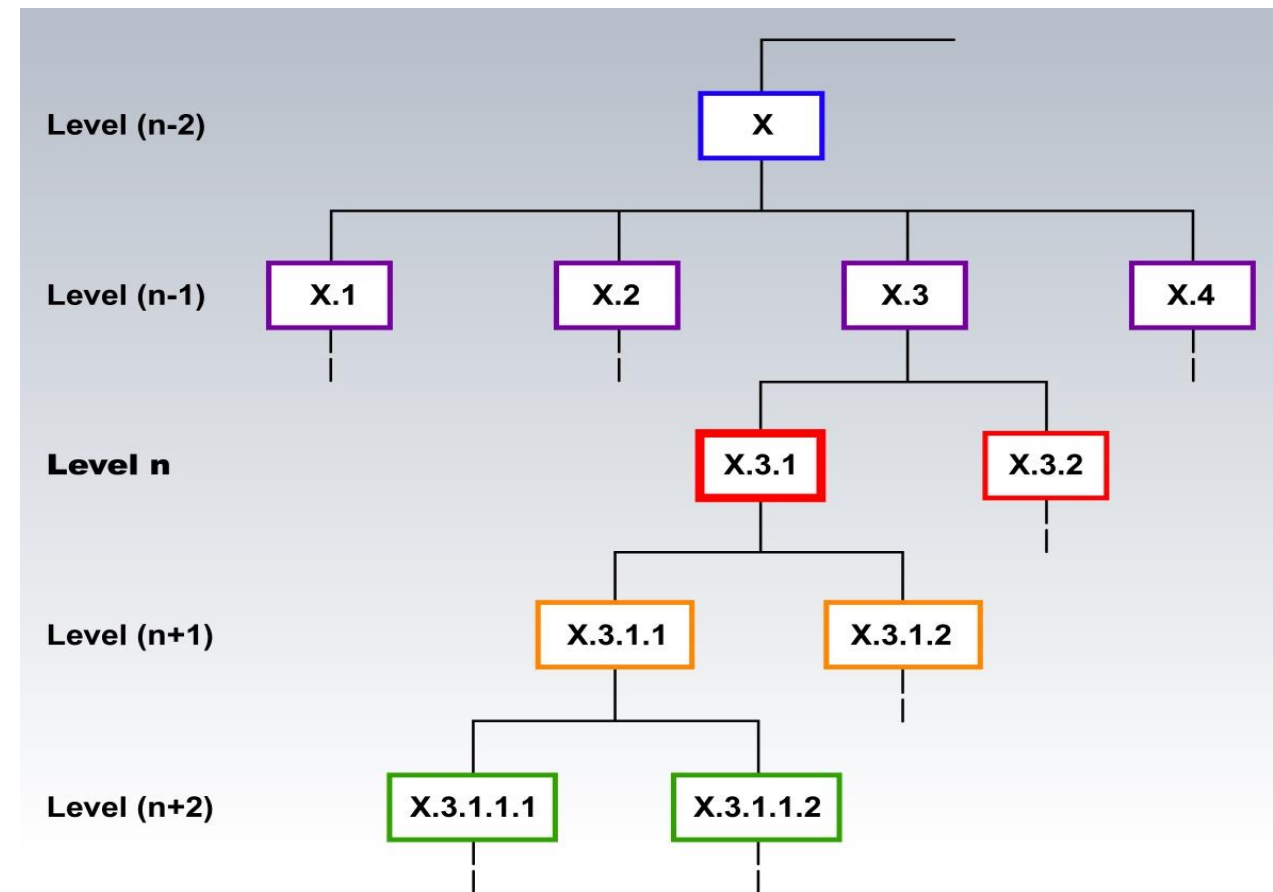


- Phoenix was a robotic spacecraft on a space exploration mission on Mars. The Phoenix lander landed on Mars on May 25, 2008.
 - Mission scientists used instruments aboard the lander to assess the local habitability and to research the history of water there.
- *Wikipedia*



- Mars Science Laboratory (MSL) is a robotic space probe mission to Mars launched by NASA on November 26, 2011, which successfully landed Curiosity, a Mars rover, in Gale Crater
 - The objectives include investigating Mars' habitability, studying its climate and geology, and collecting data for a human mission to Mars.
- *Wikipedia*

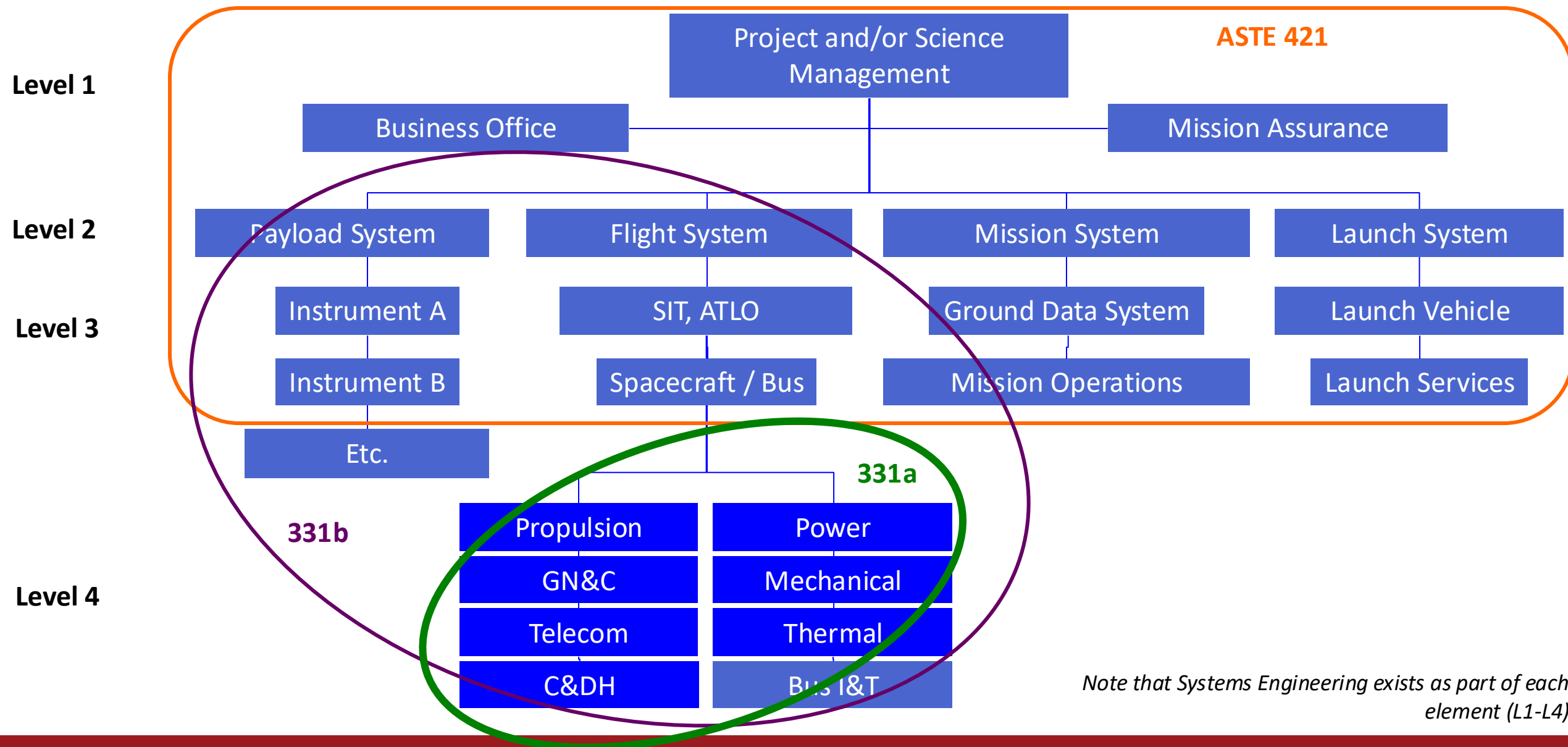
- **A system...**
 - Serves to satisfy stakeholder requirements
 - Is a hierarchy of subsystems, sub-sub-systems, etc.
 - May have super-systems, super-super-systems, etc.
- **The definition of a system (including functions interfaces, & lower-level subsystems) is functional decomposition**
- **System design addresses the architecture of components, interfaces, and constraints of the end-to-end system**
- **System development consists of all the activities to produce and integrate the resulting product**



Let X.3.1 be the system of interest. Then its subsystems are X.3.1.1 and X.3.1.2, and its sub-subsystems are X.3.1.1.1 and X.3.1.1.2. Also, X.3 is its supersystem, and X is its super-supersystem.

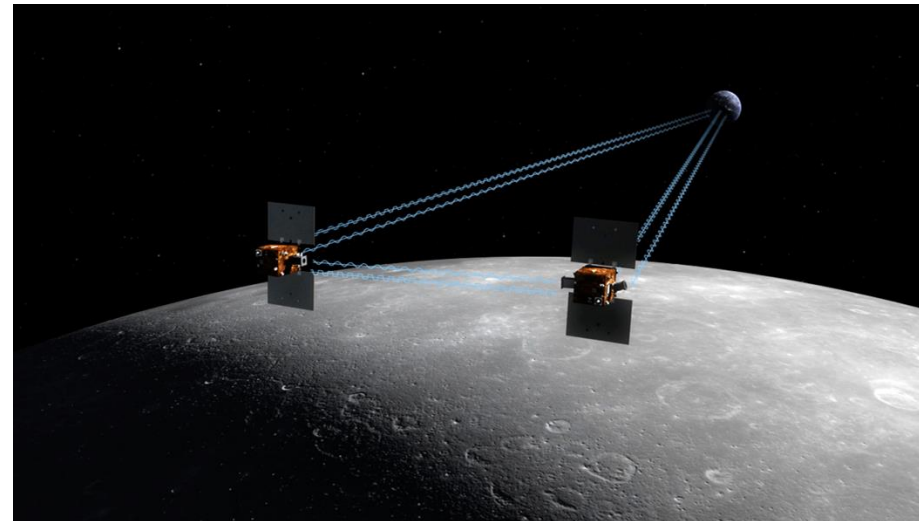
Source: System Engineering Overview, A. Ruskin

Typical Project Organization



PART 2: GRAIL CONCEPTUAL DESIGN

The following is a walk-through of the GRAIL early mission concept



The Gravity Recovery and Interior Laboratory (GRAIL) was a lunar science mission in NASA's Discovery Program which used high-quality gravitational field mapping of the Moon to determine its interior structure. The two small spacecraft were launched on 9/10/2011 aboard a Delta II LV. After their 3-4 month cruise, they arrived at their lunar orbits 25 hours apart (12/31/11 & 1/1/12). The two spacecraft impacted the Lunar surface on December 17, 2012.

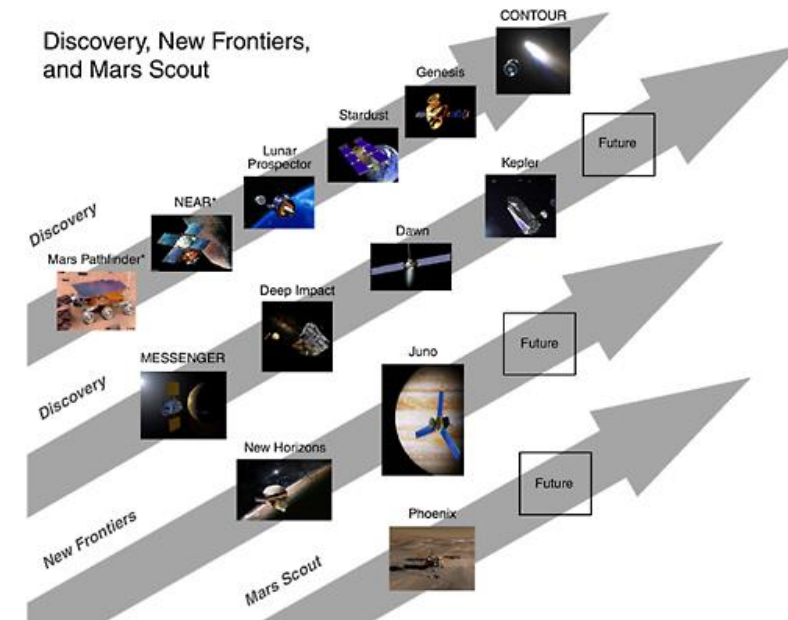
• Background

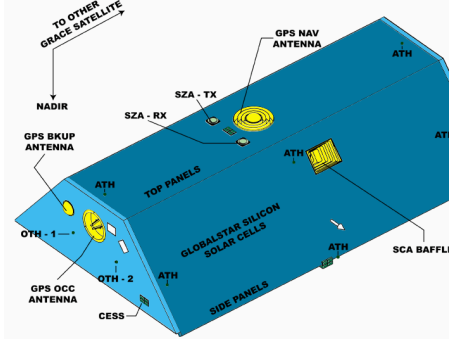
- In 2009, NASA HQ released an Announcement of Opportunity (AO) for Discovery program missions
- The Discovery Program selects Science-led space mission investigations under a not-to-exceed cost cap that address NASA's planetary science goals
 - These are defined in "Decadal Studies" and NASA exploration roadmaps
- Several NASA centers and other organizations responded to this call with more than 20 different concepts of varying levels of risk & cost
- JPL proposed several concepts, including the GRail mission that was eventually selected

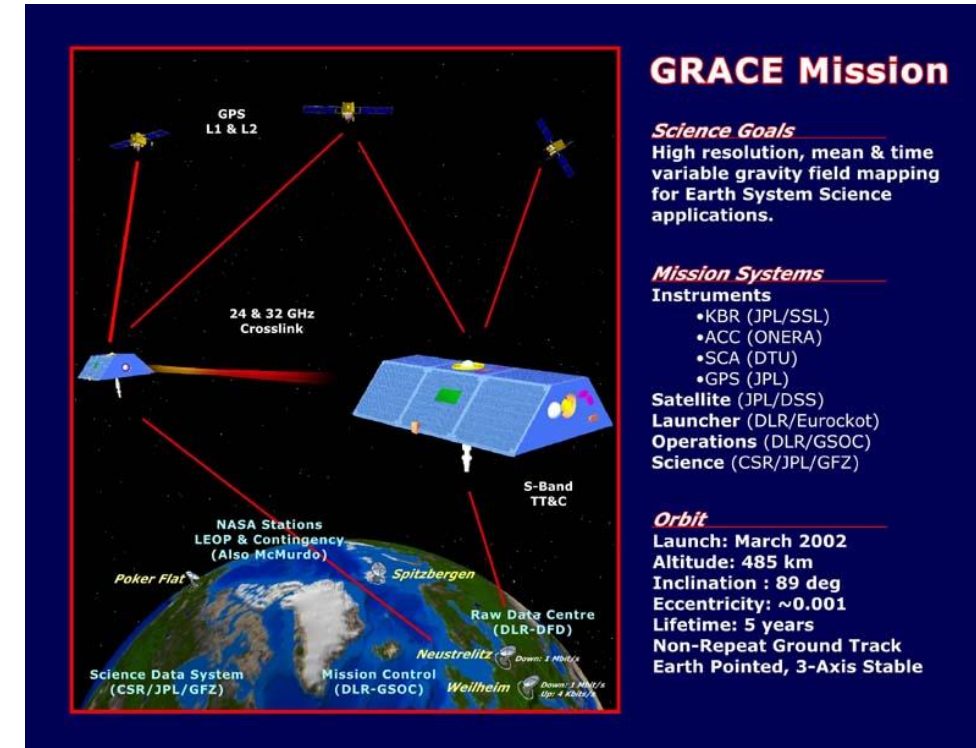
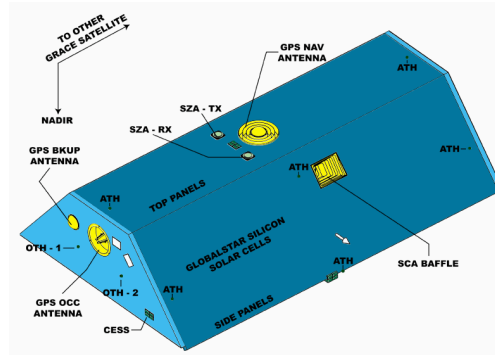
• Study Objectives and Program Requirements

- Initial GRail Study Objectives
 - Determine concept feasibility of a lunar gravity-mapping mission
 - Target a lower-risk, lower cost mission given increasingly risk-averse selections by NASA HQ
 - Coordinate with Maria Zuber (leading planetary geologist & MIT department chair)
- Constraints
 - Minimize risk (eg, no new technologies and ensure robust margins for mass, power, etc.)
 - Cost cap of $\leq \$425$ million, launch no earlier than January 2012
 - Schedule: 4-5 years for development, 3-6 months operations

NASA Uses AOs to Solicit Mission Proposals



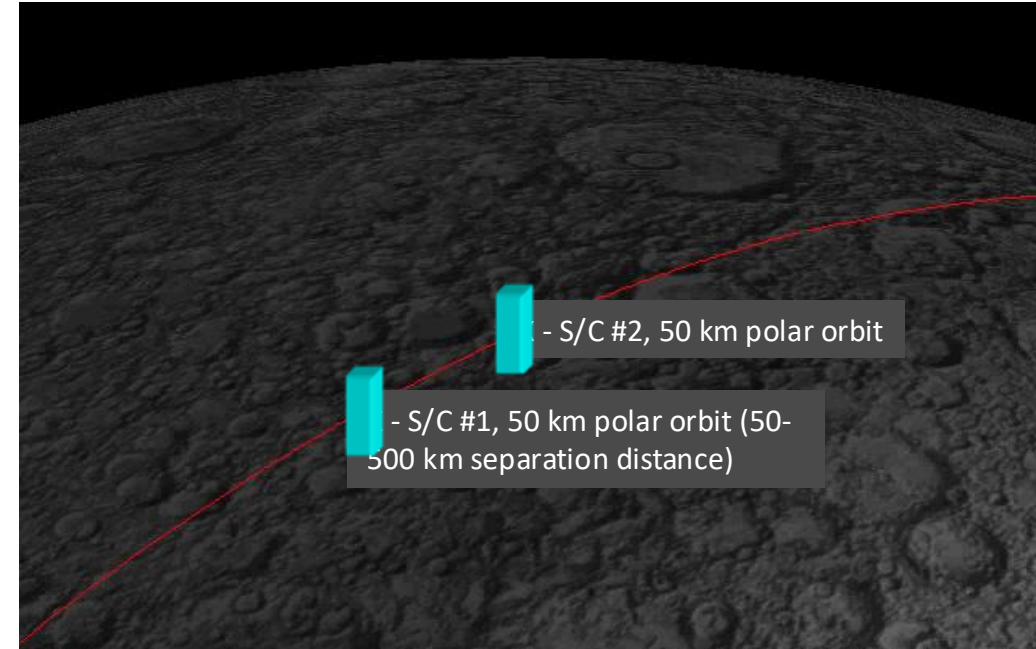
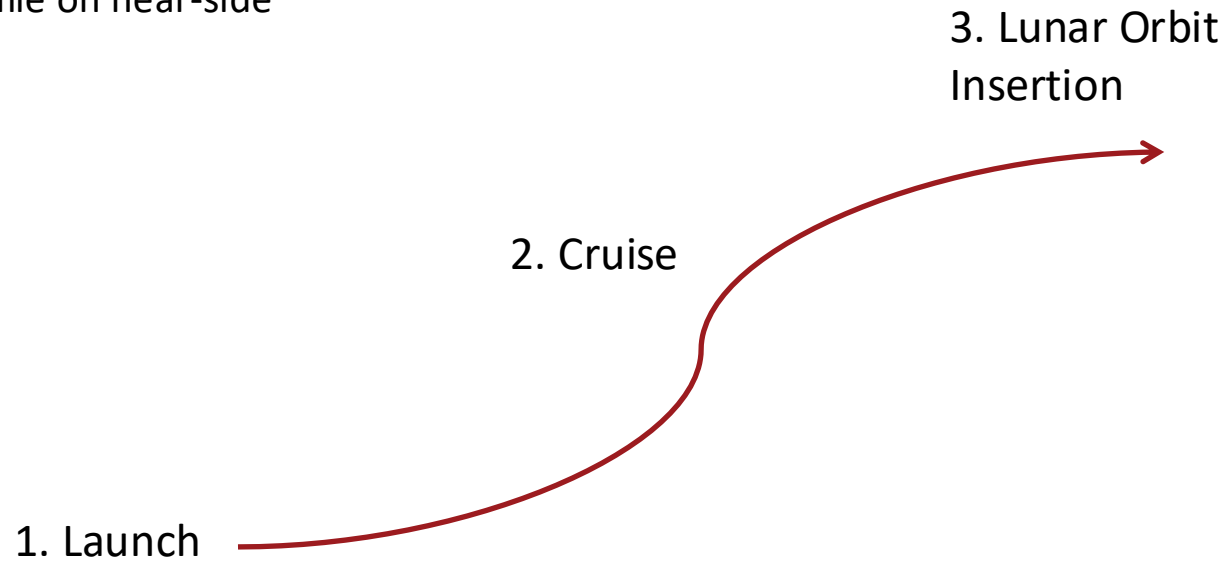
- **The Initial Idea**
 - GRACE at the moon
- 
- The diagram illustrates the Lunar Reconnaissance Orbiter (LRO) in a 3D perspective view. It is a blue, elongated spacecraft with various instruments and antennas labeled. At the front, there is a 'GPS NAV ANTENNA' and a 'GPS BKUP ANTENNA'. A 'GPS OCC ANTENNA' is located on the side. The main body features 'GLOBALSTAR SILICON SOLAR CELLS' on the 'TOP PANELS' and 'SIDE PANELS'. Other labels include 'NADIR', 'TO OTHER GRACE SATELLITE', 'SZA - TX', 'SZA - RX', 'ATH', 'OTH - 1', 'OTH - 2', 'CESS', and 'SCA BAFFLE'.
- **Science Objectives (from Decadal Survey)**
 - Determine the structure of the lunar interior, from crust to core.
 - Advance understanding of the thermal evolution of the Moon.
 - Extend knowledge gained on the internal structure and thermal evolution of the Moon to other terrestrial planets.
 - **Additional Objective**
 - Produce detailed lunar gravity field required for any future scientific or exploration objective (for Constellation program, 2005-2009).



Basic Concept of Operations

Communication with Earth

- Deep Space Network (likely 34 m antenna) while on near-side



4. Science Operations at the Moon (~90 days)
5. Decommission (crash onto surface)

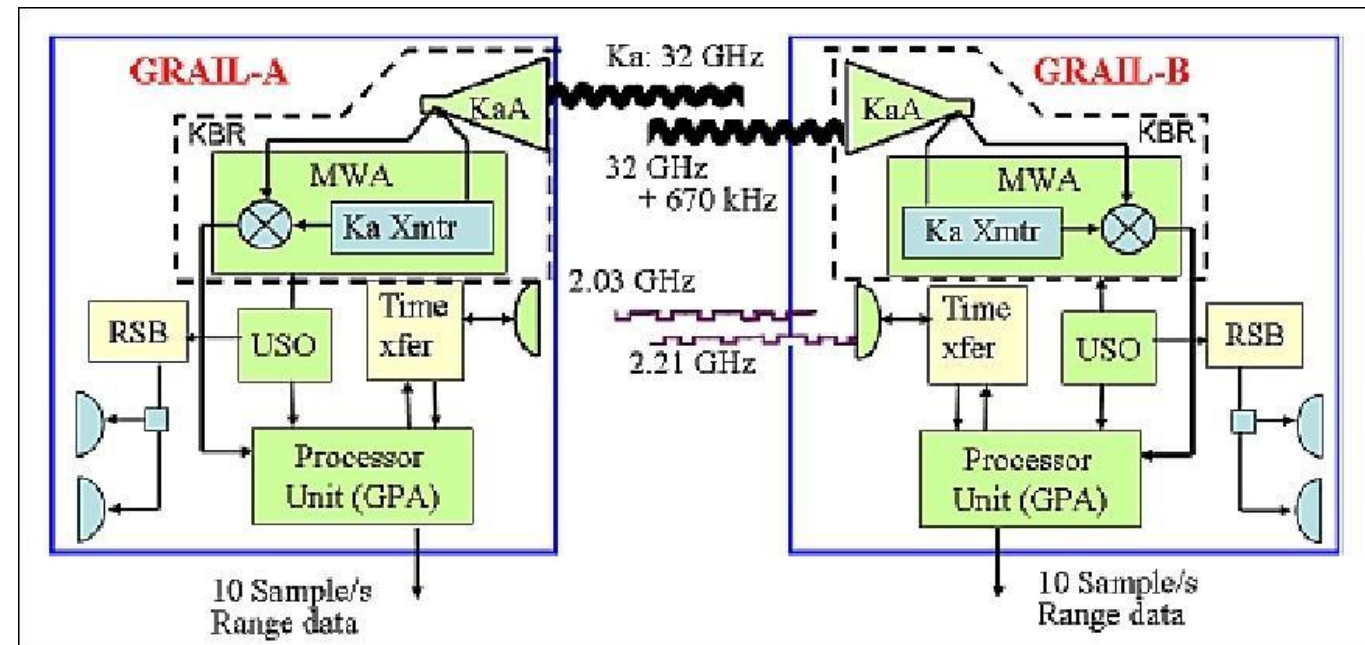
Initial Questions

- Payload requirements?
- Mission design (ie, trajectory) for lunar orbit insertion (LOI)?
- Spacecraft design?



- Two satellites will each carry an enhanced Ka-Band Ranging & Instrument Assembly based on the GRACE KBR instrument developed by JPL. It uses a Ka-Band radio frequency to precisely determine satellite-to-satellite range, and the two spacecraft must point at each other. This instrument is under concept development, and its expected requirements are listed below:
 - Instrument: Enhanced Ka-Band Ranging & Instrument Assembly
 - Mass: 15 kg (10 kg + 50% contingency)
 - Power: 15 W (10 W + 50% contingency)
 - Pointing Requirements: Control- 0.25 deg, Knowledge- 200 arcsec
- Instrument Data Rate: 10 MB/day

- Two spacecraft
- Accommodates 15 kg, 15 W payload
- 3-axis stabilized
- Pointing & data rate should not be drivers, but must be accommodated



<https://directory.eoportal.org/web/eoportal/satellite-missions/g/grail>

Destination

- 50 x 50 km polar lunar orbit
- 90-deg inclination

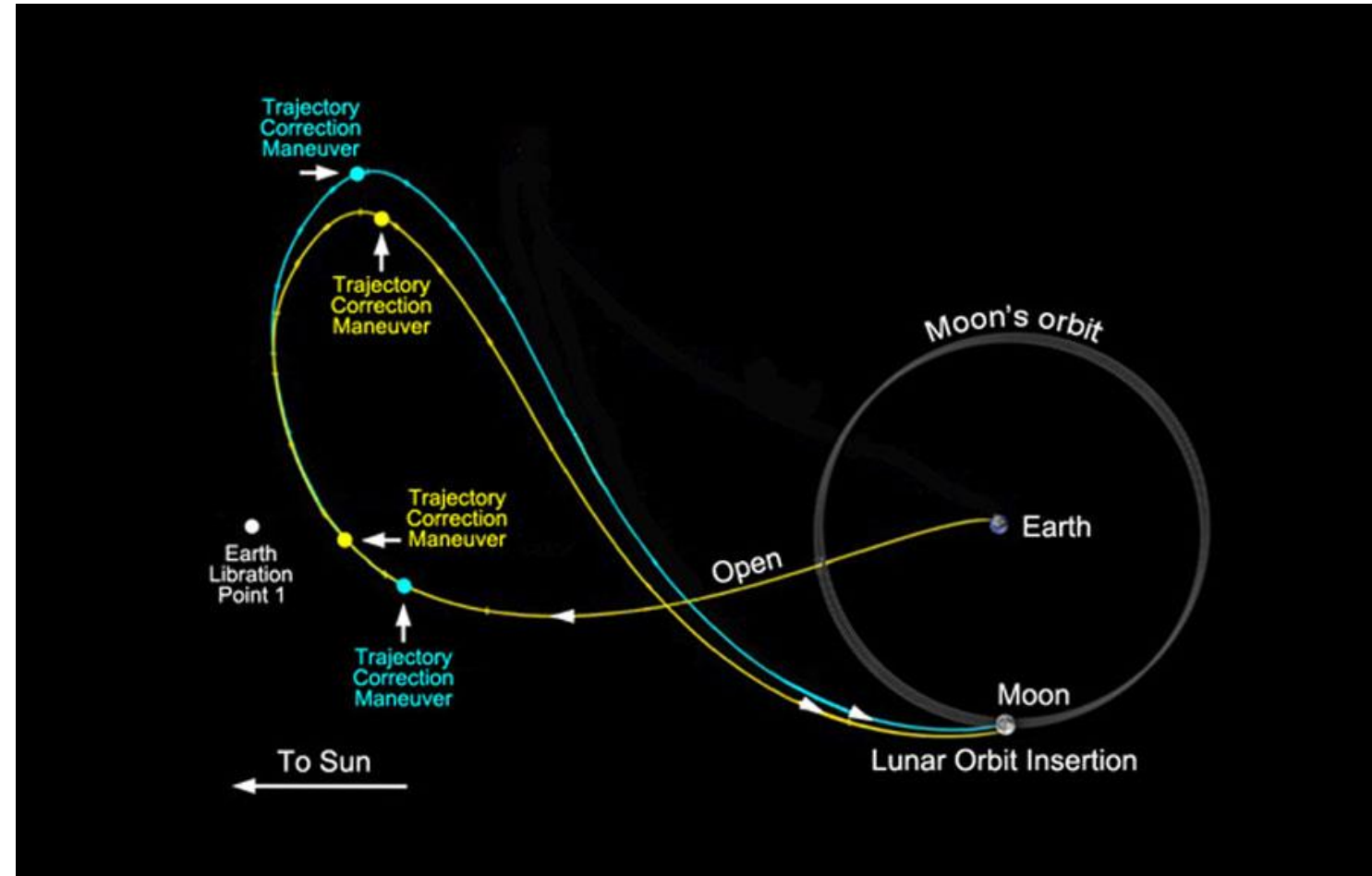
Two Trades

1. Earth Escape vs. Staging

1. Option 1: Earth Escape
 - LV insertion to lunar rendezvous
2. Option 2: Staging
 - LV insertion to 185 x 43,800 km
 - 5 chemical burns to spiral out to lunar rendezvous

2. Direct Insertion vs. Low Energy

- Option 1: Direct Insertion
 - Lunar orbit insertion to 100 x 8,000 km
 - 5 chemical burns to spiral in to 50 x 50 km lunar orbit
- Option 2: Low Energy (using Lagrangian points)
 - LV insertion into low energy orbit
 - Lunar orbit insertion to 100 x 8,000 km
 - 5 chemical burns to spiral in to 50 x 50 km lunar orbit



		<u>Option 1</u> Earth- escape + Direct	<u>Option 2</u> Earth- escape + Low Energy	<u>Option 3</u> Staging + Direct	<u>Option 4</u> Staging + Low Energy	Comments
Launch Vehicle Capability	LV Injection	C3 = -2.0	C3 = -0.7	185 x 30,000 km	185 x 30,000 km	
	Taurus 3113	445 kg	435 kg	637 kg	637 kg	Assumes 63" fairing (~60 kg less performance for larger fairing)
	Delta 2326-9.5	605 kg	585 kg	920 kg	920 kg	From KSC LV site
Earth- escape to Lunar Rendezvous Trajectory	LV Correction	~30 m/s	~30 m/s	~15 m/s	~15 m/s	
	Escape Burn (burns include gravity losses)	0 m/s	30 m/s	813 m/s	870 m/s	Burn may be broken into segments, depending on prop system
Lunar Orbit Insertion to 100 x 100 km orbit	Lunar Targeting	5 m/s	5 m/s	5 m/s	5 m/s	
	Lunar Orbit Insertion	825 m/s	697 m/s	825 m/s	697 m/s	May be broken into segments, depending on prop system
Descent to 50 x 50 km Polar Orbit	Lunar Descent Burn	23 m/s	23 m/s	23 m/s	23 m/s	
Mass Delivered to Final Orbit	Taurus 3113	299 kg	305 kg	298 kg	308 kg	Assumes lsp of 226 s for all delta-V burns
	Delta 2326-9.5	406 kg	411 kg	431 kg	445 kg	
Other Info	Total Delta-V	883 m/s	785 m/s	1681 m/s	1610 m/s	
	Total Flight Time (cruise)	~6 days	~90 days	~15 days	~99 days	Flight times are approximate
	Max Earth Range	~400,000 km	1,500,000 km	~400,000 km	1,500,000 km	

Driving Requirements

- ?

		Option 1 Earth- escape + Direct	Option 2 Earth- escape + Low Energy	Option 3 Staging + Direct	Option 4 Staging + Low Energy	Comments
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Driving Requirements

- Delta-V of 800-1600 m/s
- Flight time 5 to 100 days
- Max Earth range:
 - 400k to 1,500k km

Questions

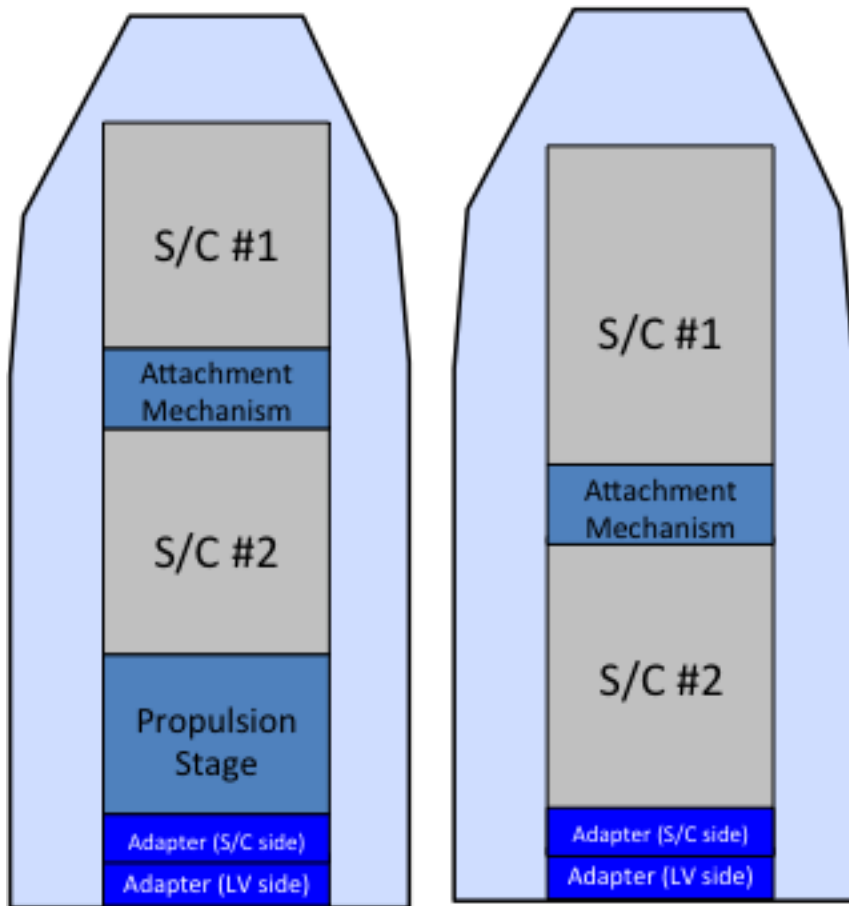
- Propulsion System
 - Integrated prop system or separate?
 - Biprop vs. monoprop?

100-200 kg for
each (considered
small spacecraft)

Propulsion Trades

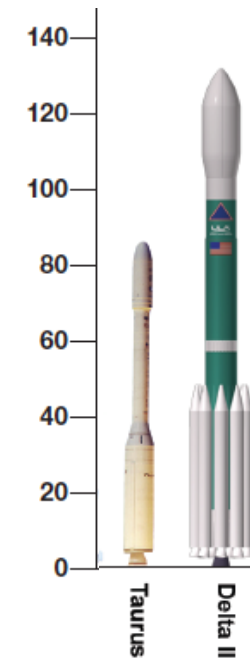
Propulsion System Questions

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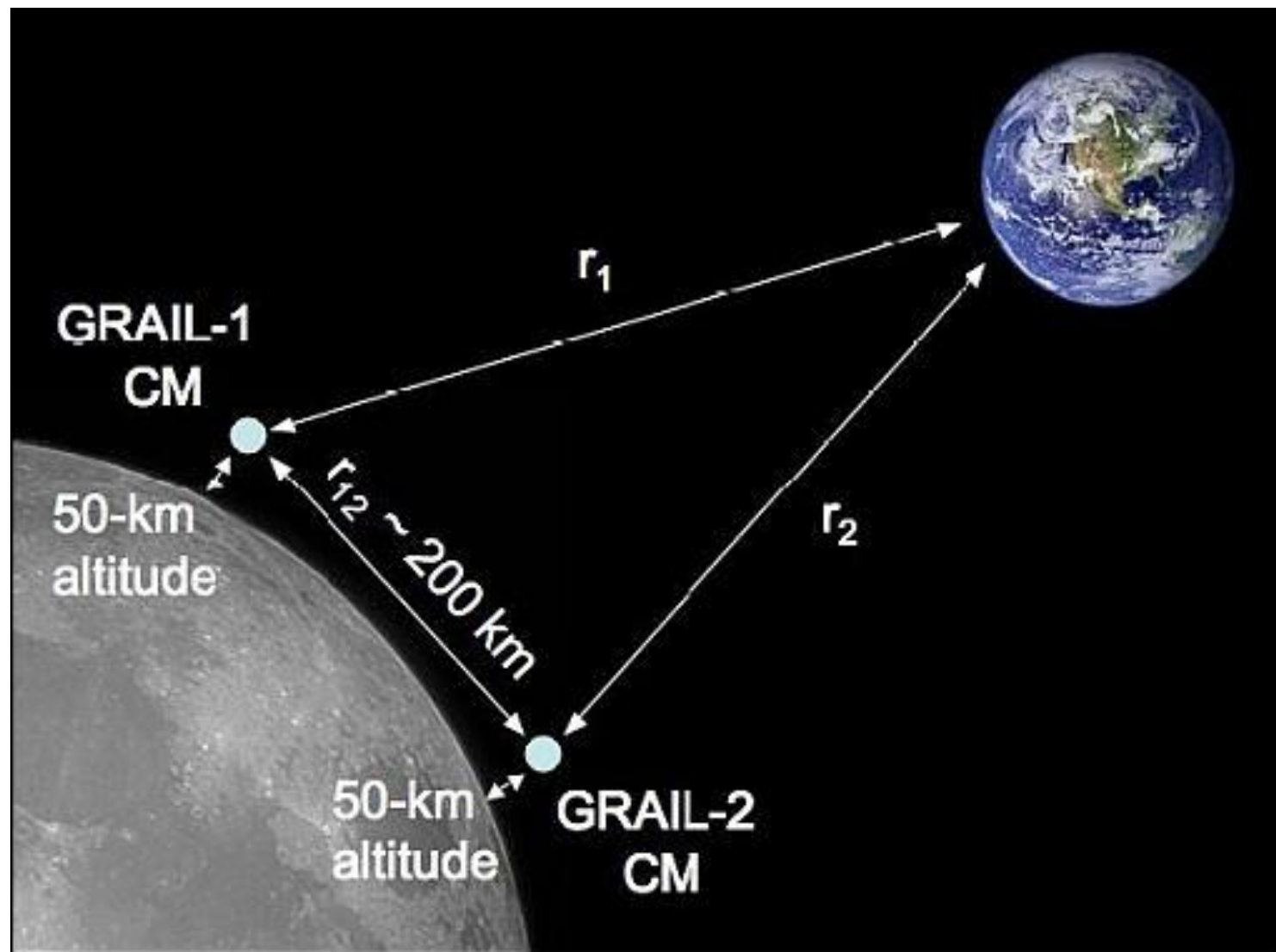


Launch Vehicle?

- Taurus 3113
 - Delta II
- (\$50 M difference at the time)



Orbit Maintenance & Operations



Orbit Parameters

- 2 Spacecraft
- 50 x 50 km polar orbit
- Period: 113.0 minutes
- 50 km separation distance

Mission Operations

- Duration
 - 3 months
- Tracking (on near-side)
 - Continuous
- Downlink
 - ~10 MB/day
- 34-m DSN

Updated Mission Requirements

Mission Design

- **Launch Date:** January 2012 (flexible)
- **Mission Duration:** 5-100 day cruise, 3 months prime operations
- **Trajectory & ΔV :** See prior charts
- **Operations:** 58 m/s for maintenance
- **Launch Vehicle:** Taurus 3113 (includes Star 37FM with 63" fairing)

Rationale

AO

Trajectory Analysis

Trajectory Analysis

Orbital Analysis

Largest Taurus LV

Flight System

- **Flight System:** TBD (100-150 kg dry mass, w/monoprop vs. biprop system)
- **Redundancy:** Single string with & functional redundancy
- **Stabilization:** 3-axis with reaction wheels (?) & propulsion system
- **Prop Stage:** TBD (Monoprop or biprop stage to deliver s/c)
- **Telecom:** 64 kbps S-band?

Short Duration Mission

Lunar Orbit Maintenance

Trajectory Analysis

Link Budget Analysis

Programmatic

- **Schedule:**
 - Phase A: 7 months
 - Phase B: 11 months
 - Phase C/D: 36 months
- **Reserves:** 30% for mass, power, & cost (unless noted otherwise)

Rule of Thumb

Rule of Thumb

Survey Criteria

- Mass < 150 kg
- Pointing Knowledge < 350 arcsec
- ACS is 3-axis with reaction wheels and no magnetic torquers
- Includes propulsion system
- Primarily RSDO spacecraft considered, which excludes most foreign s/c

Results

- Most small satellites are designed for Earth orbit, using magnetic torquers rather than RCS propulsion systems
- Spectrum-Astro SA-200S is close in mass and carries a monoprop propulsion system
- Additional non-RSDO spacecraft should be further investigated

RSDO Spacecraft		Dry Mass	Data Rate	Pointing Knowledge	ACS	Propulsion
Organization	Model	kg	kbps	arcsec	type	type
Surrey	SNAP	6.5	100	3600	3-axis, mom-bias, mag tq	Liq. Butane
Surrey	Microsat-70	44.7	2048	1800	3-axis, zero-mom, mag tq	-
Orbital	PicoStar	52.7	100	3600	spinner	-
Orbital	MicroStar	58.6	2000	2880	3-axis, zero-mom, mag tq	-
Spectrum Astro	SA-200B	90	2500	317	3-axis, zero-mom, mag tq	-
Orbital	MiniStar	100	115.2	540	spinner	-
Spectrum Astro	SA-200S	129	2500	2.8	3-axis, zero-mom	Monoprop
Orbital	LeoStar-2	166	2000	108	3-axis, zero-mom, mag tq	-
TRW	T100	184.1	200	694	3-axis, zero-mom	Monoprop
Ball Aerospace	BCP 600	203	17000	12	3-axis, zero-mom, mag tq	-
Surrey	Minisat-400	206.7	2048	72	3-axis, zero-mom	Cold Gas
TRW	T200A	242.4	1400	324	3-axis, zero-mom	Monoprop
TRW	T200B	278	2048	2	3-axis, zero-mom	Monoprop
Swales	EO-SB	332	2000	36	3-axis, zero-mom, mag tq	-
Spectrum Astro	SA-200HP	354	80,000	0.5	3-axis, zero-mom	Monoprop
Astrium	FlexBus	383	1000	2063	3-axis, zero-mom	Cold Gas
Lockheed Martin	LM 900	492	TBD	10	3-axis, zero-mom	Monoprop
Orbital	StarBus	566.3	1000	390	3-axis, zero-mom	Monoprop
Orbital	MidStar	580	1024	10	3-axis, zero-mom, mag tq	-
Ball Aerospace	BCP 2000	608	320000	3	3-axis, zero-mom	Monoprop
TRW	T-310	718	1024	43	3-axis, zero-mom	Biprop

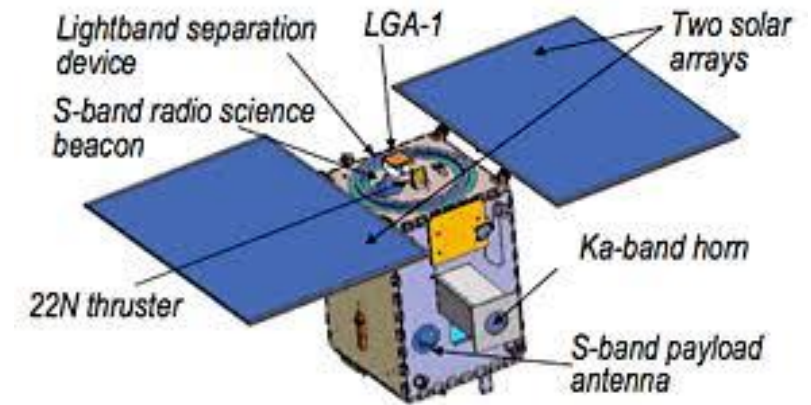
Additional Non-RSDO S/C

(not considered)

S/C Buses (50 to 150 kg)	Mass (kg)	Power (W)
Oersted, Terma, DEN	50	26
Sloshsat, Dutch Space, NET	58	35
Constella, SSTL, UK	60	120
IRIS, Fuchs Gruppe, BEL	67	44
STRV, DERA, UK	70	60
RS300, Ball, USA	74	120
Myriade, Astrium, CNES, FR	74	80
Bird, GER	78	40
TechSat21, MicroSat, USA	82	433
XSS-11, LM, USA	85	100
TE, MicroSat, USA	90	120
ODIN, SSC, SWE	90	160
PROBA, Verhaert, BEL	94	30
MMB-100, SpaceDev, 100	100	80
PegaStar, Orbital, USA	100	150
MITA, Fuchs Gruppe, ITA	120	100
SciSat, Bristol, CA	150	70

Note: GRACE used Astrium's Flexbus

- Based on a preliminary design, a Spacecraft Bus Request for Information (RFI) was released to solicit bus concepts
- RFI included
 - Trajectory/Propulsion options (this trade was still open)
 - Payload specifications
 - Flight system requirements
- Result was that Lockheed was chosen with a bus derived from its prior XSS-11 spacecraft
- Result from initial study was that the concept was feasible and likely budget would be \$310-\$350 M (very close to the final cost)



PART 3: OVERVIEW OF SPACECRAFT SUBSYSTEMS

• Function

- Provides thrust for both trajectory design maneuvers and (often) attitude control
- The primary driver is typically the trajectory and corresponding ΔV budget.
- Across all subsystems, components are typically selected as those that meet the minimum requirements with the lowest mass, power, & cost.

• Common Components

- Chemical Propulsion (Mono/Biprop)
 - Fuel tanks
 - Thrusters (often different types tailored for small/large maneuvers)
 - Integration/etc. hardware
- Solid rocket motors (SRMs)
- Electric Propulsion Systems

• GRAIL Example

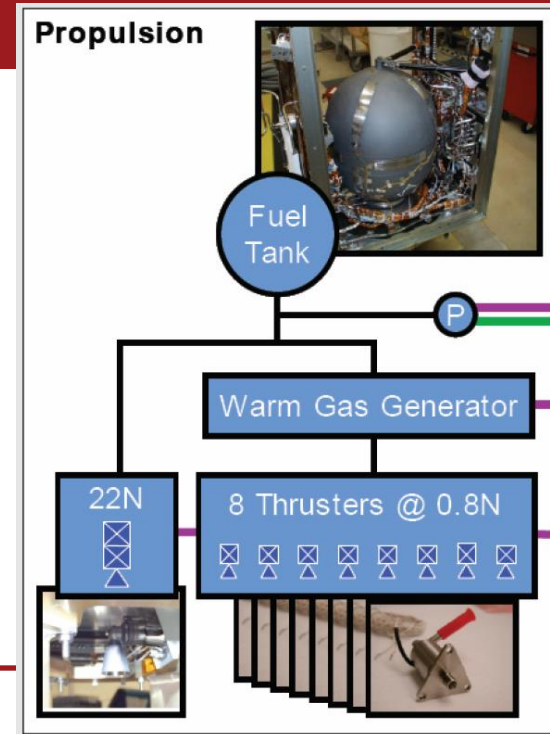
- 1 22-N thruster
- 8 0.8 N thrusters
- Warm gas generator
- Fuel tank
- Misc. hardware
- Total = ~15 kg

• Key Trades & Analyses

- Propulsion type (solid, mono/bi, electric)
- Thruster & fuel tank sizing
- Heritage from prior systems

• Key Parameters

- Subsystem mass, power, cost
- Thruster Isp, locations, cant angles
- Propellant load & margins



Attitude Control System (ACS)

• Function

- Provide attitude control, knowledge, and stability to support spacecraft pointing
- Primary drivers for pointing are:
 - Science observations (instrument FOVs)
 - Trajectory course maneuvers (thrust)
 - Telecom (uplink/downlink to Earth)
 - Thermal (wrt sun, Earth, etc.)
- Also called GNC, ADCS, AOCS, etc.

• Common Components

- Sensors
 - Sun sensors, magnetometers
 - Gyros, GPS receivers
 - Star trackers
- Control Effectors
 - Reaction or momentum wheels
 - Control moment gyros
 - Magnetic torquers
 - Reaction control thrusters (see propulsion)

• GRAIL Example

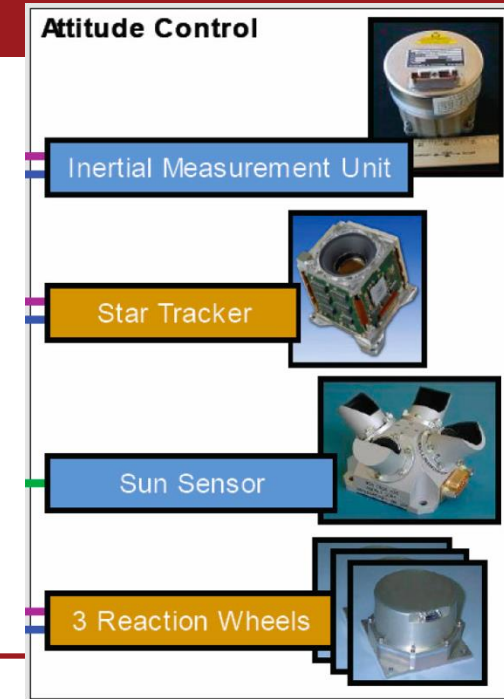
- 1 star tracker
- 3 reaction wheels
- 1 inertial measurement unit (IMU)
- 1 sun sensor
- Total = ~5 kg

• Key Trades & Analyses

- 3-axis control, spin, gravity gradient...
 - Reaction wheels vs. thrusters, etc.
- Control/knowledge/stability analyses and error budgets
- Heritage from prior systems

• Key Parameters

- Mass, power, and cost
- Pointing control, knowledge, stability/jitter



Function

- Provides communication to/from the spacecraft. Typically with the Earth, but can include other spacecraft.
- Occasionally used for radio science
- Primary driver is typically spacecraft-Earth distance

Common Components

- Radio/transponder
- Amplifier
- Antennas
 - High-gain (HGA)
 - Medium-gain (MGA)
 - Low-gain (LGA)

GRAIL Example

- S-band transponder and diplexer
- Low-gain antenna (LGA)
- Misc. hardware
- Total = ~5 kg

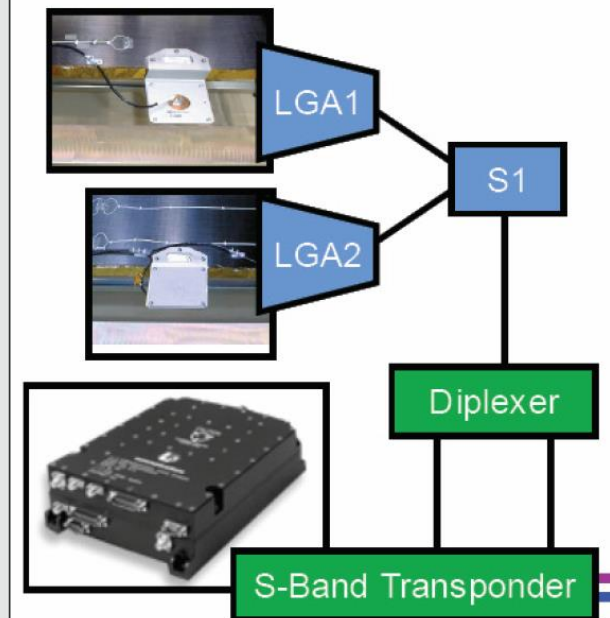
Key Trades & Analyses

- Fixed vs. articulated
- Data rate & volume analysis
- Link budget & antenna sizing
- Radio-band and power level
- Heritage from prior systems

Key Parameters

- Mass, power, and cost
- Transmission power, data, & margin

Telecom



Control & Data Handling (C&DH)

• Function

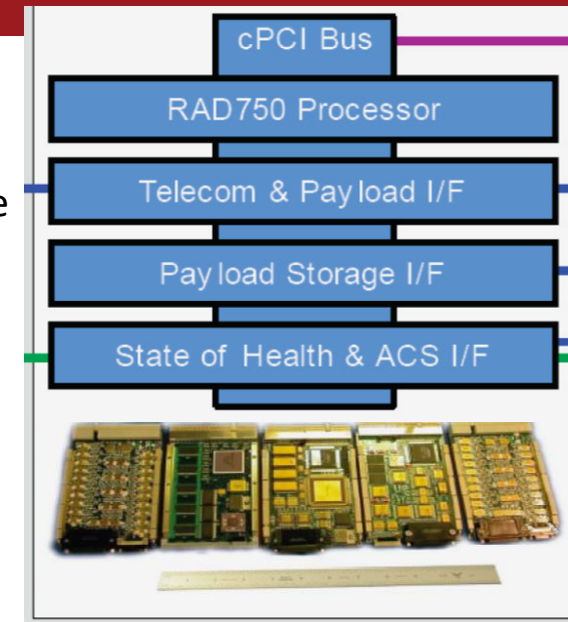
- Provide spacecraft command and control, along with internal data handling. This includes FSW with boot/reset functionality.
- Primary drivers are the redundancy, interfaces, & complexity of the downstream components (eg, payload)
- Also called Flight Computer

• Common Components

- Electronics box/chassis
 - Processor (eg, Rad750)
 - Payload interface cards
 - Data interface cards
 - Memory
- Depending on complexity, the above functions may be in a single box or multiple boxes

• GRAIL Example

- Rad750 processor
- I/F cards, converters, backplane
- Chassis assembly
- Total = ~5 kg



• Key Trades & Analyses

- Functional block diagram
 - Interface types (1553, LVDS, RS-422, etc.)
- Onboard memory storage & CPU %
- Radiation shielding analysis
- Heritage from prior systems

• Key Parameters

- Mass, power, & cost

• Function

- Provide electrical power and distribution throughout the spacecraft
- Primary drivers are typically
 - Sun-spacecraft distance
 - Power profiles during science & telecom operations
 - Launch for battery sizing
 - Redundancy
- Often paired with C&DH under “Avionics”

• Common Components

- Generators
 - Solar arrays
 - Radioisotope thermoelectric generators (RTGs)
- Storage Devices
 - Fuel cell, primary battery, secondary battery
- Power processors
 - Regulate voltage and distribute current

• GRAIL Example

- Power electronics (interface cards, converters)
- 1 30 Ahr battery
- 2 solar arrays
- Total = ~30 kg

• Key Trades & Analyses

- Solar arrays vs. RTGs
- Battery type (eg, primary vs. secondary)
- Array & battery sizing
- Power equip. list (PEL) & power analysis
- Heritage from prior systems

• Key Parameters

- System power loads, profiles, & margins
- Worst-case depth of discharge

Electronics



Power



30 A-hr Li-Ion Battery



2 Solar Array s

• Function

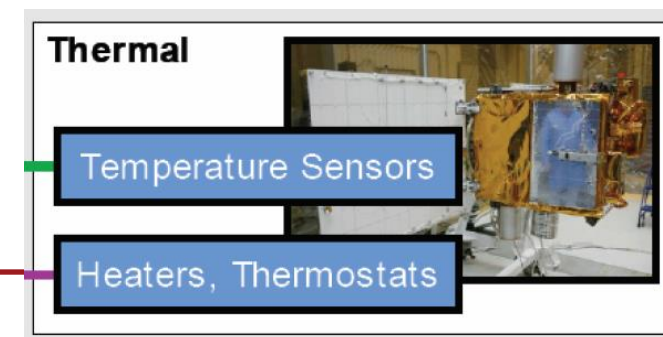
- Provide appropriate thermal control throughout the spacecraft
- Primary driver is the overall thermal model that includes temperatures from:
 - Avionics (usually room temperature)
 - Instruments (eg, cold FPAs or hot sample analyses)
 - Batteries, mechanisms, propulsion
- All components are required to stay within their qualification temperature limits

• Common Components

- Surface coatings & treatments
 - Films, paints, MLI
- Conductors and insulators
- Heat transfer regulators
 - Heat pipes, louvers, phase-change devices, heat switches

• GRAIL Example

- Paint, films
- Multi-layer insulation (MLI)
- Instrument thermal mass plate
- Heaters, thermostats, sensors, etc.
- Total = ~ 10 kg



• Key Trades & Analyses

- Active versus passive
- Thermal analysis
- Radiator design, locations, & margins

• Key Parameters

- Mass, power, & cost
- Updated attitude profile

• Function

- Provides structure to support spacecraft and payload, along with all mechanisms and separation systems
- Primary drivers are:
 - Spacecraft type (orbiter vs. rover)
 - Payload (camera vs. sampling system)
 - Overall dry & wet mass
 - Articulated arrays, antennas, instruments
- Also called Structure & Mechanisms, and often includes the harness (ie, wiring)

• Common Components

- Primary structure
 - Core structure, panels, etc.
- Secondary structure
 - Tank support, fittings, etc.
 - Payload support, booms
- Payload elements (wheels, sample return canisters, etc.); can also go under Payload
- Mechanisms
- Launch vehicle and spacecraft adapters

• GRAIL Example

- 4 side panels
- Propulsion & payload panels
- Tank support, fittings...
- Solar array deployment mech.
- LV separation system
- Total = ~35 kg



• Key Trades & Analyses

- Configuration, CAD, & FOV design
- Loads and stiffness analysis
- Master equipment list
- Mechanism, torque, etc. analyses

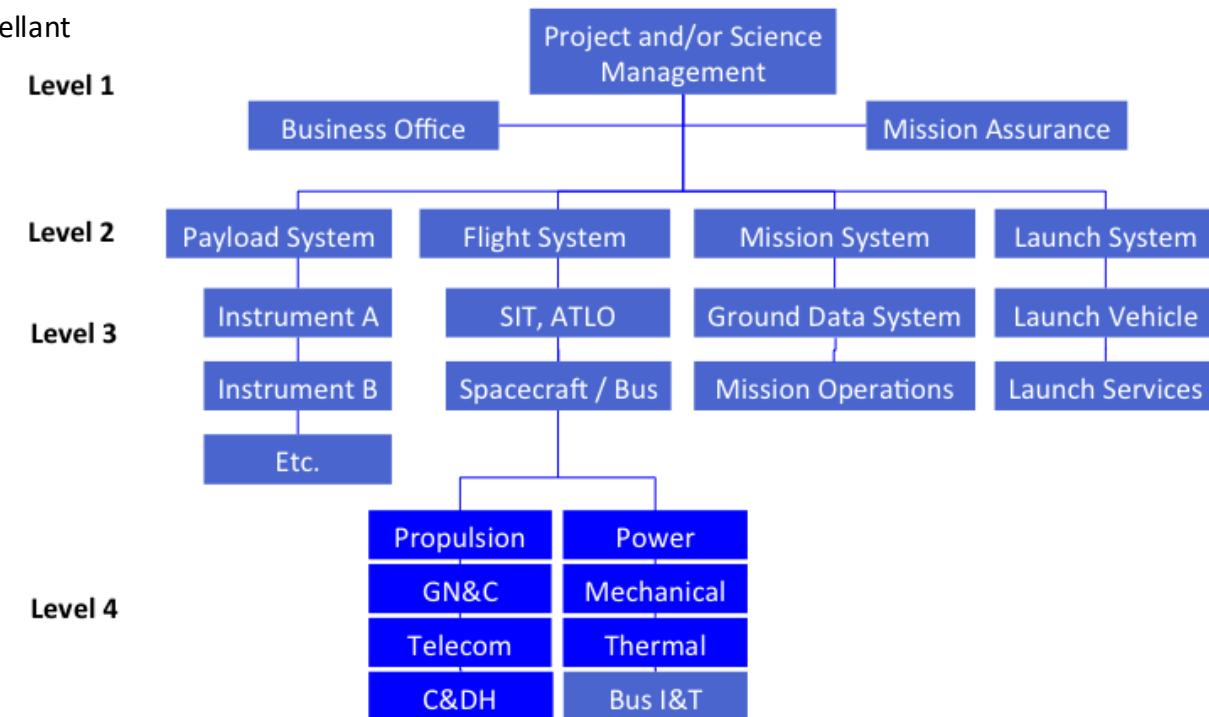
• Key Parameters

- Mass, power, & cost

PART 4: MISSION REQUIREMENTS FLOW-DOWN

Requirements Introduction

- A *requirement* is “a singular documented physical or functional need that a particular design, product or process aims to satisfy.” – [wikipedia](https://en.wikipedia.org/wiki/Requirement)
- Requirements are used in the Aerospace industry to:
 - Map a system hierarchy to a corresponding set of required functions to ensure the Project can achieve the mission objectives:
 - For example:
 - At Level 1: The Mission shall achieve an Earth orbit that supports the science requirements
 - At Level 2: The Project shall place an Orbiter in a 500 km, geocentric Earth orbit
 - At Level 3: The Spacecraft shall provide a total ΔV capability of 1,000 m/s
 - At Level 4: The propellant tank shall store 100 kg of propellant
- Requirements are written via shall statements
 - The XYZ shall...



Note that, by design, there is typically a strong correlation between organization (people), requirements (tasks), and functional product deliveries (s/c & subsystems)

Requirements Introduction

- **Requirements are also used in the Aerospace industry to:**
 - Protect capability and margins across the flight system, particularly across interfaces within (or external) to the organization
 - For example:
 - The Project systems team knows that the spacecraft will provide 1,000 m/s
 - » Vs. the actual need of 900 m/s
 - The Spacecraft team knows that the Propulsion will provide 100 kg
 - » Vs. actual need of 90 kg
 - Note that requirements form the basis of many contracts between the customer and a contractor (or between a contractor and a subcontractor). Payment is often contingent on meeting these requirements.
 - Ensure institutional best practices
 - For example:
 - The Spacecraft shall perform DSN compatibility testing use the Flight System prior to launch

Questions about Requirements

- **Why do we write requirements?**

- Understand the end-to-end design prior to hardware
- Communicate between customer & supplier

Iterating on the end-to-end design to produce a set of detailed requirements generally occurs in the first $\frac{1}{2}$ to $\frac{1}{3}$ of the project (eg, 2-3 years).

- **What do we write requirements for**

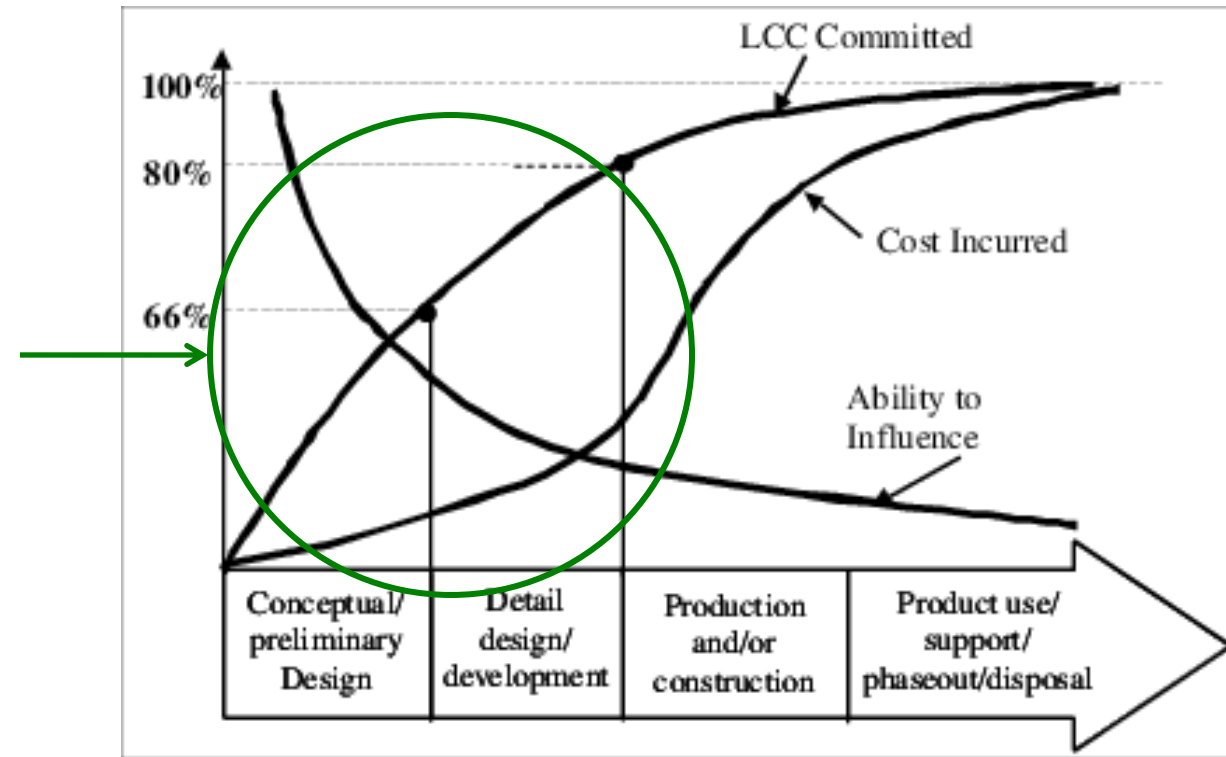
- Typically for new developments
- For existing hardware, requirements are written on its operation

- **When do we stop writing requirements?**

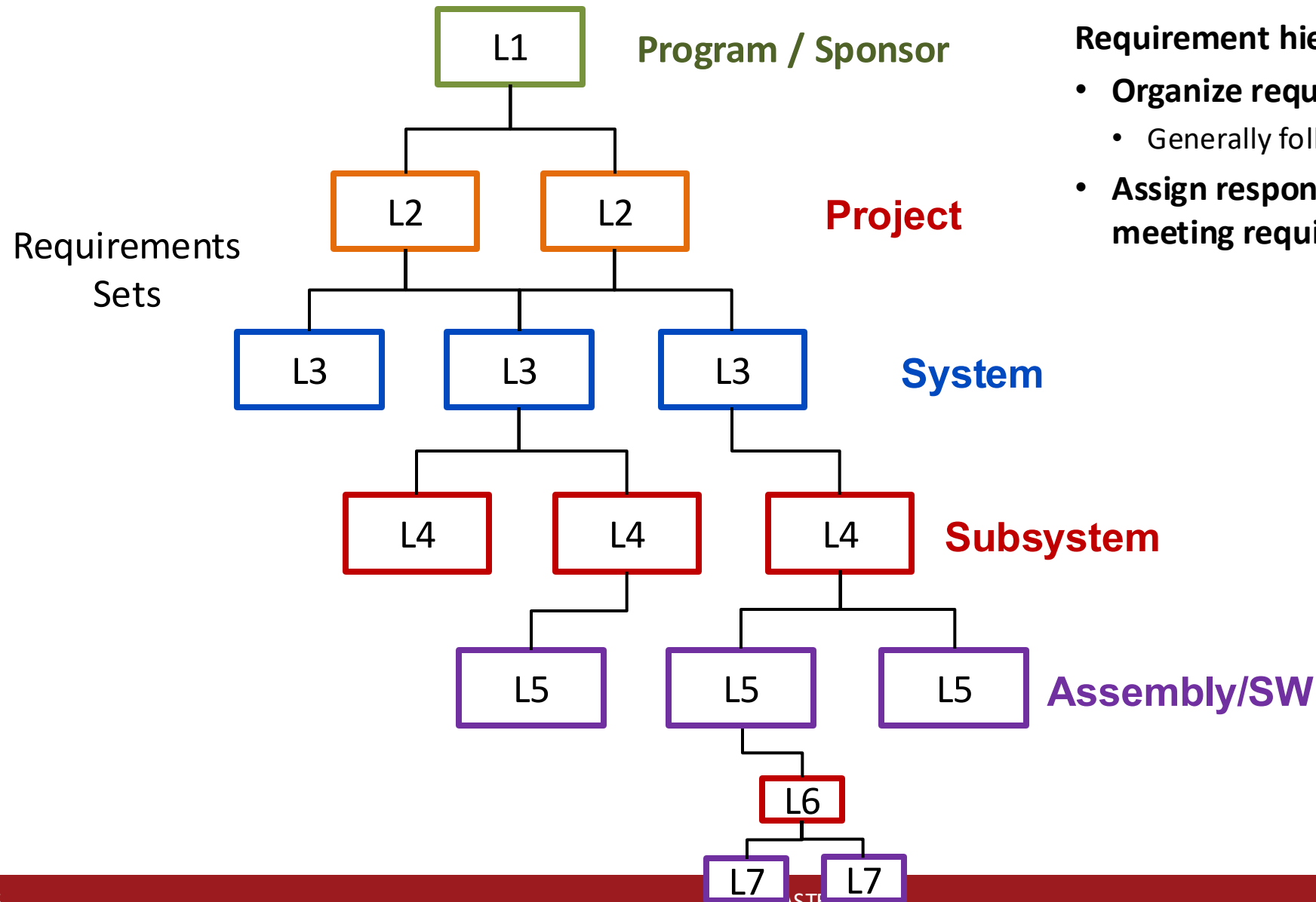
- When there are no more new development items in the system description

- **Why do we verify requirements?**

- Ensure that the system meet its specified requirements
- Understand the degree to which it meets the requirements



Requirements Hierarchy



Requirement hierarchy exists to:

- **Organize requirements work**
 - Generally follows the product architecture,
- **Assign responsibility for generating or meeting requirements to elements or teams**

Requirements Level Definitions

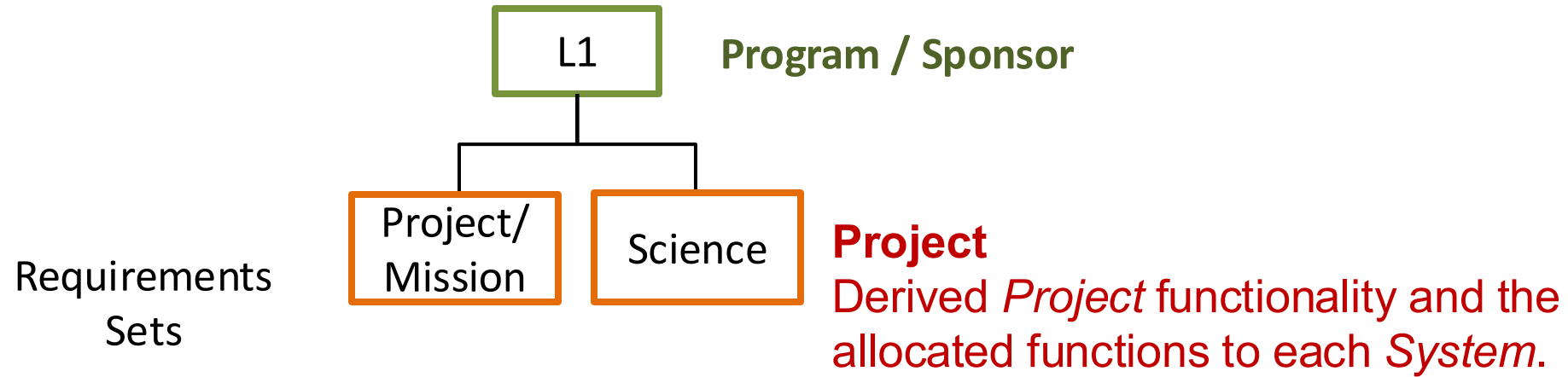
L1

Program / Sponsor

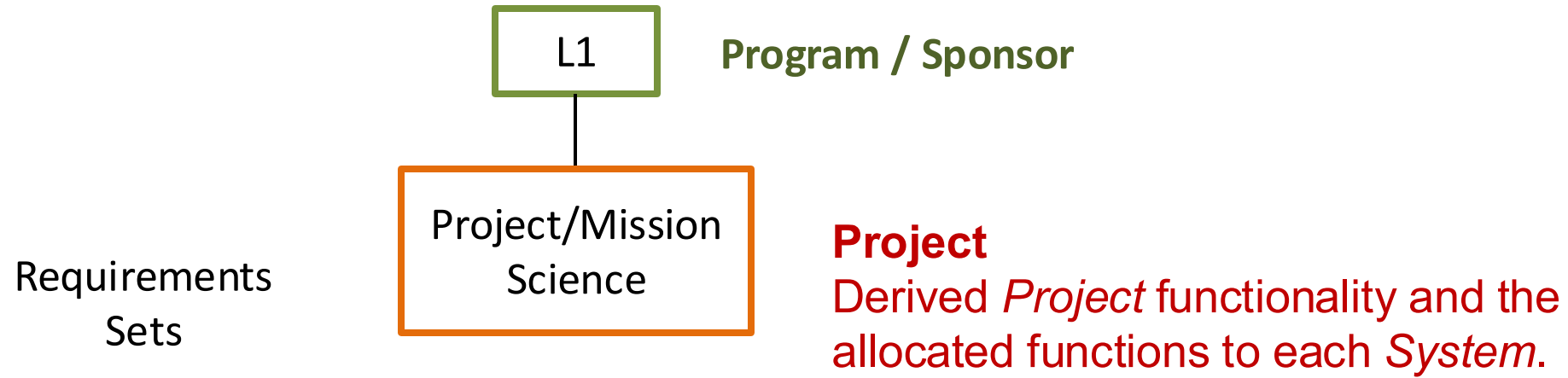
*Sponsoring organization or program-derived
and allocated requirements on the Project.*

Requirements
Sets

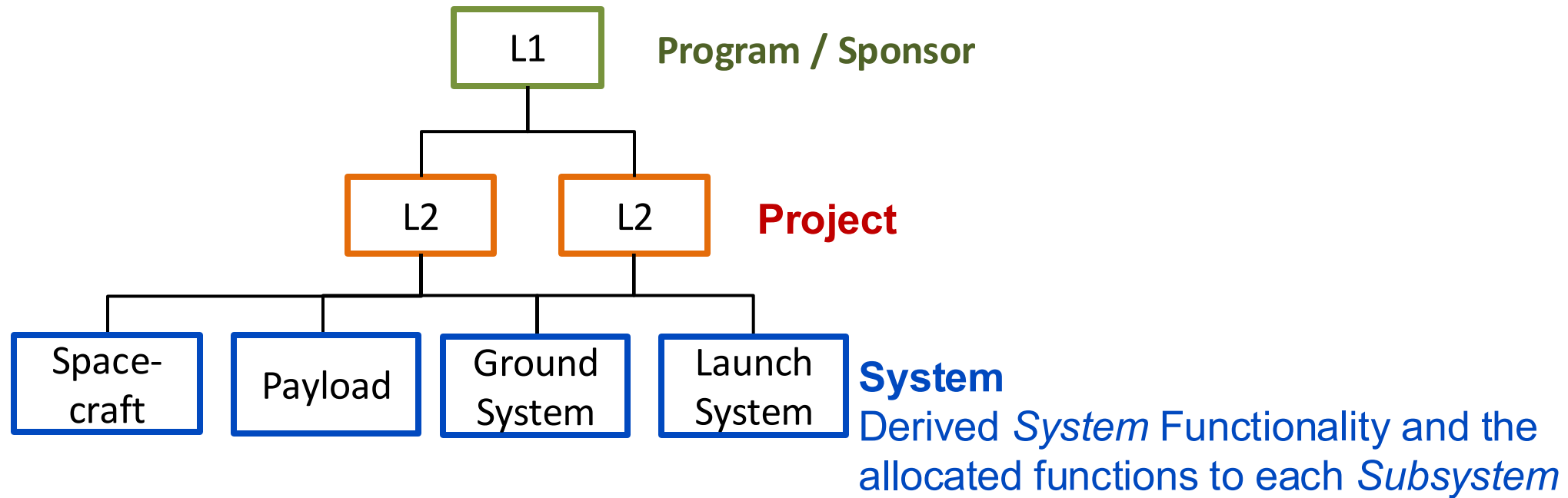
Requirements Level Definitions



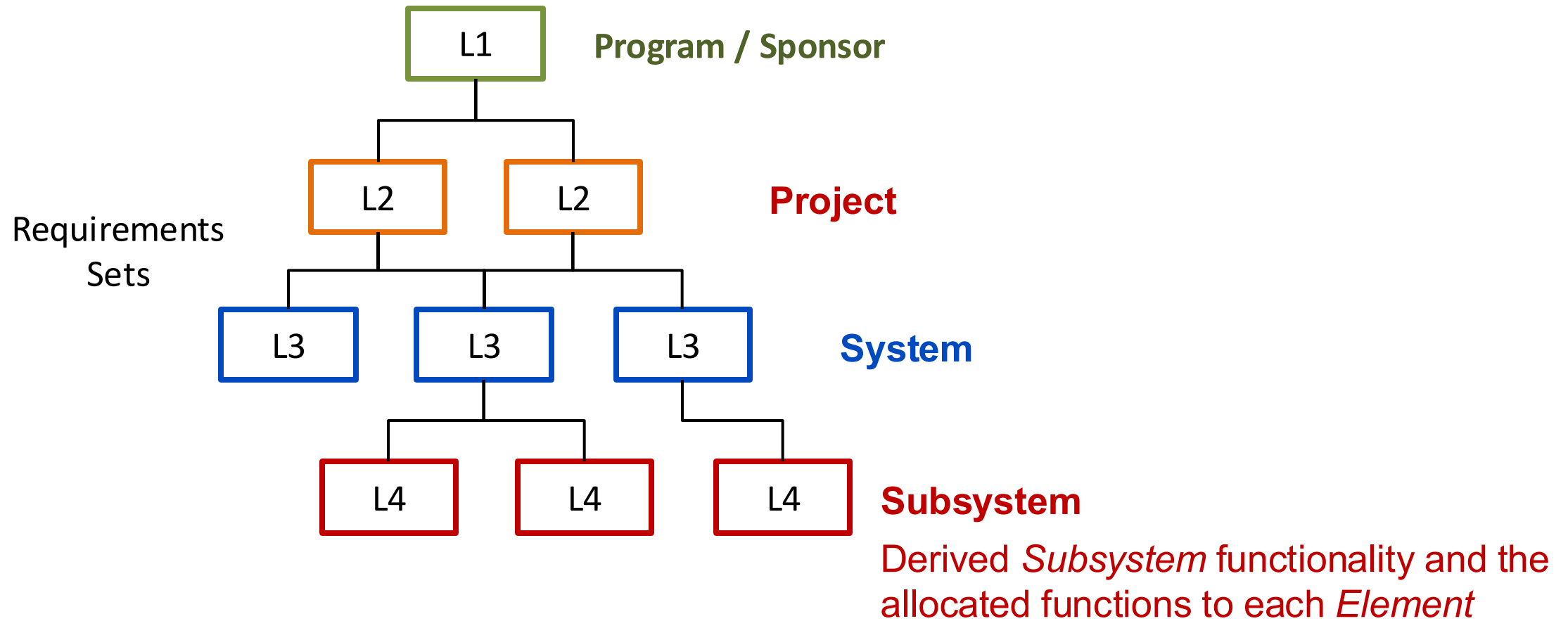
Requirements Level Definitions



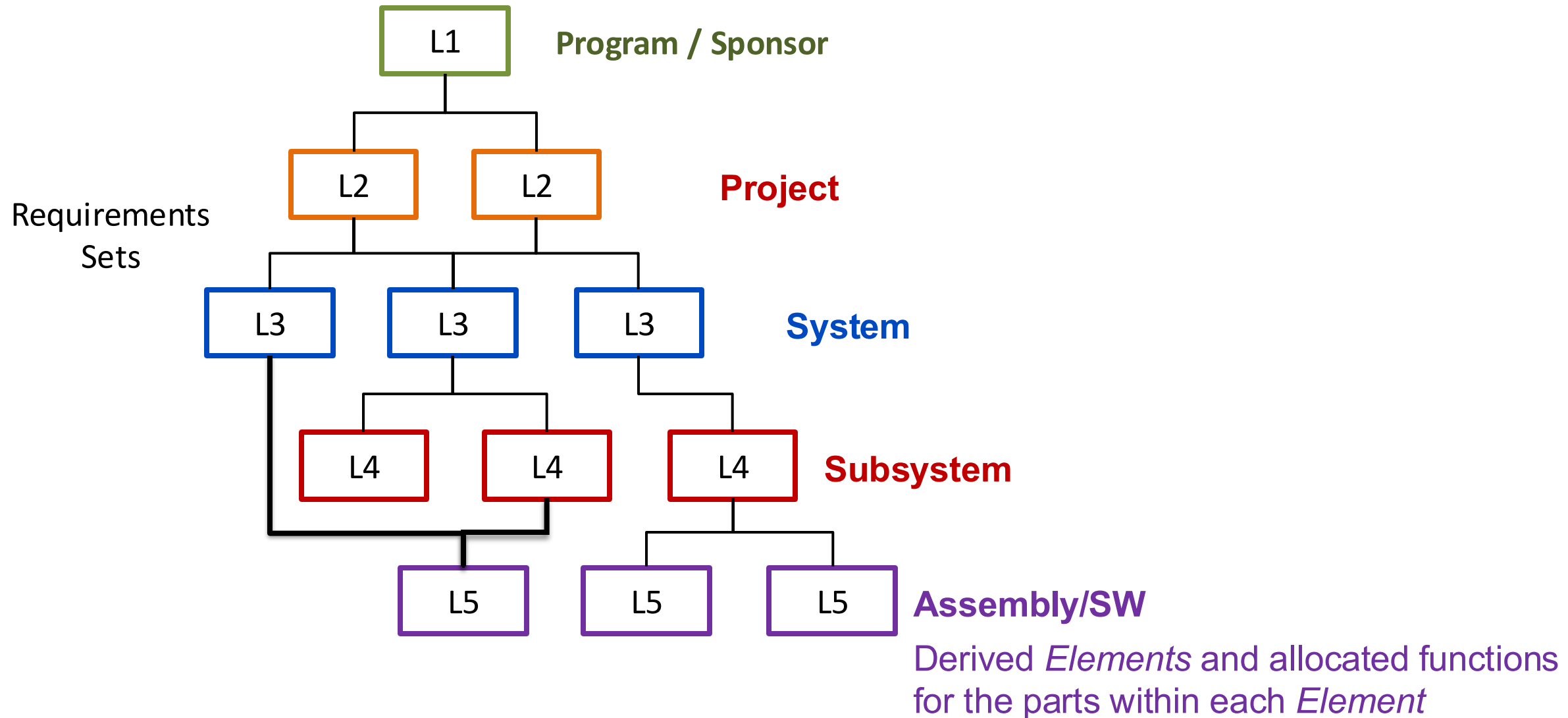
Requirements Level Definitions



Requirements Level Definitions

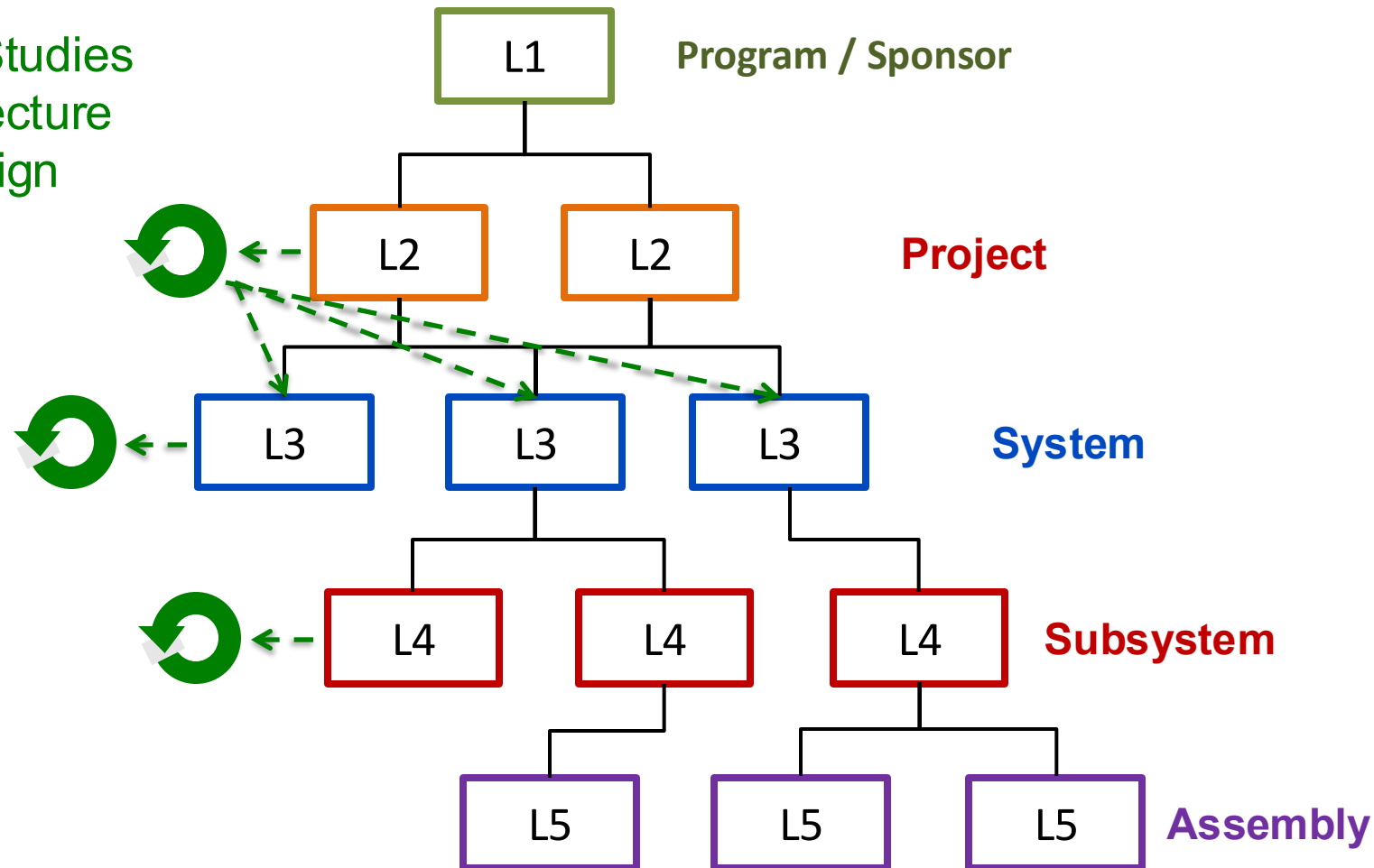


Requirements Level Definitions



Deriving Next Level Requirements

Trade Studies
Architecture
Design



In order to produce high-quality L1 requirements, you really need to understand L3/L4s

Example Requirements Flow Down

